SIDEWALL TECHNOLOGIES

# OmniPilot

## Complete Report

Personal AI Assistant + Hardware Platform
Software, Research, Market Intelligence & Hardware Strategy

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## PART 7: EXECUTION ROADMAP

PART 1

# The Vision

## What is OmniPilot?

**THINK OF IT THIS WAY**

Imagine a personal secretary who sits in every meeting with you, remembers every conversation you've ever had, knows your business finances, tracks your investments, manages your customer relationships, and can act on your behalf across seven different business tools -- all through voice commands. That secretary never forgets, never sleeps, and costs less than Rs 500/month. That's OmniPilot.

OmniPilot is a **voice-first, local-first personal AI companion** that listens, remembers, and acts. Unlike Siri or Alexa that forget everything after each query, OmniPilot builds a persistent memory of your conversations, meetings, decisions, and tasks. Unlike Omi or Limitless that are standalone gadgets, OmniPilot is deeply wired into the Sidewall product portfolio -- seven business tools that span advertising, billing, communication, trading, interior design, and software development.

The project started in April 2026 as a macOS desktop companion. Within two weeks, a working prototype was running: always-listening via Silero VAD, transcribing with Whisper, remembering in SQLite, reasoning with Ollama's Qwen3 model, and speaking back with Apple's voice synthesis. No cloud. No subscription. Everything local.

## The Evolution: Software to Platform

OmniPilot is not a single product -- it's a **platform strategy** with three expanding rings:

| RING | WHAT | STATUS | TIMELINE |
|---|---|---|---|
| **Ring 1: Software** | macOS desktop companion + WhatsApp bot + Flutter mobile app | v0.3 working (desktop) | Now - Month 3 |
| **Ring 2: Hardware Pendant** | BLE audio pendant (Omi competitor) for always-on capture | Research complete | Month 4 - 8 |
| **Ring 3: Vertical Hardware** | AI vision device for blind users + drone voice control system | Deep research done | Month 6 - 18 |

**STRATEGIC INSIGHT**

Each ring feeds the next. The software platform validates the AI pipeline (VAD + STT + LLM + memory). The pendant proves the hardware form factor. The vertical devices (blind assistive, drones) take the same pipeline into high-impact, high-margin markets where India has massive unmet demand.

## OmniPilot by the Numbers

| 11 | 5 | 28 | 3 | 7 |
|---|---|---|---|---|
| SOURCE FILES | SERVICES | RESEARCH CHARTS | HARDWARE VERTICALS | PRODUCT INTEGRATIONS |

## Key Achievements So Far

| ACHIEVEMENT | DETAIL | STATUS |
|---|---|---|
| **Working macOS App** | Swift menu bar app with 5-tab UI (Live / Ask / Tasks / Memories / Settings) | v0.3 running |
| **Always-On Listening** | Silero VAD (neural network, 95% accuracy, 0.4% CPU) detects speech in real-time | Complete |
| **Local Speech-to-Text** | whisper.cpp with CoreML, small.en model, <500ms per chunk on M-series Mac | Complete |
| **Local LLM Reasoning** | Ollama + Qwen3 8B Q4, 20-30 tokens/sec, conversation analysis & summarization | Complete |
| **Persistent Memory** | SQLite + FTS5 full-text search + sqlite-vec semantic embeddings (384-dim) | Complete |
| **Task Scheduler** | Natural language task creation ("remind me at 3pm") + voice notifications | Complete |
| **Voice Output** | AVSpeechSynthesizer with Indian English voice (Rishi) for spoken responses | Complete |
| **WhatsApp Processing** | Voice note watcher: downloads, transcribes, stores in memory DB | Prototype |
| **Embedding Service** | all-MiniLM-L6-v2 (ONNX) for semantic search, hybrid with keyword search | Complete |

## Source Code Architecture

| MODULE | FILES | PURPOSE |
|---|---|---|
| **App/** | OmniPilotApp.swift, Pipeline.swift, VoiceOutput.swift, NotificationHelper.swift | Main app, audio-to-LLM pipeline, speech synthesis, notifications |
| **Audio/** | AudioCapture.swift, SimpleVAD.swift, WhisperBridge.swift | Microphone capture, voice activity detection, Whisper STT bridge |
| **LLM/** | OllamaClient.swift | Ollama REST API client for local LLM inference |
| **Memory/** | MemoryStore.swift | SQLite + FTS5 + sqlite-vec hybrid memory database |
| **Tasks/** | IntentParser.swift, TaskScheduler.swift, TaskStore.swift | NLU intent parsing, scheduled reminders, task persistence |
| **UI/** | QueryView.swift | 5-tab SwiftUI interface |
| **Services/** | embedding_service.py, vad_service.py, whatsapp_watcher.py, whisper_server.sh | Python microservices for ML models and WhatsApp integration |

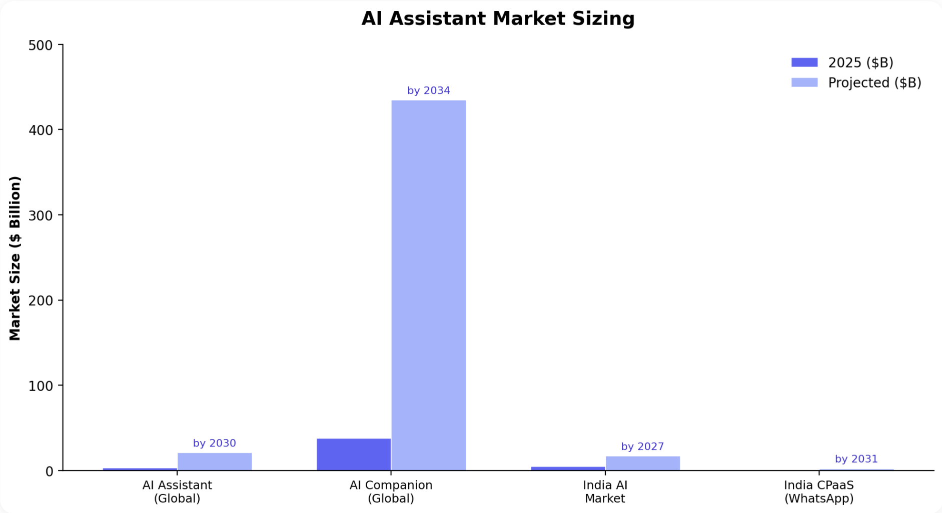
**PRIVACY PROMISE**

Zero cloud dependency. Audio, transcripts, and memories never leave the device. The entire stack -- VAD, Whisper, Ollama, SQLite, embeddings -- runs locally on any M-series Mac with 16GB RAM. Total RAM footprint: ~7GB. No API keys, no telemetry, no data collection.

PART 2

# Market & Competition

## AI Assistant Market Size



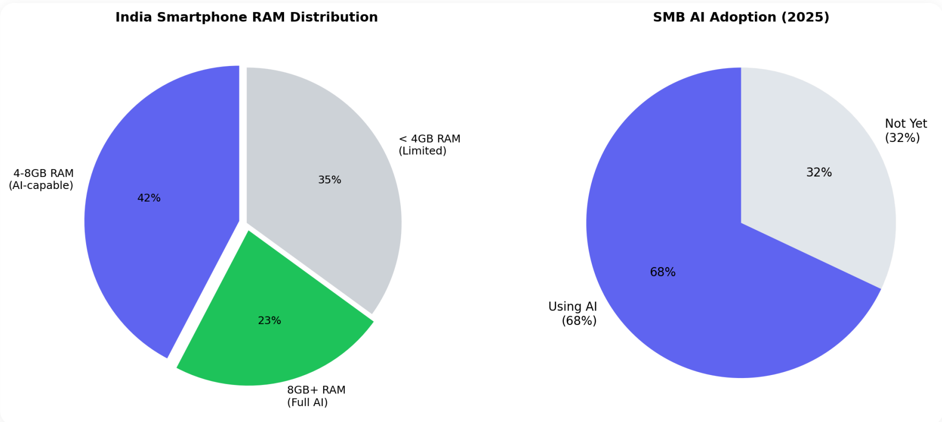

The AI assistant market is experiencing explosive growth across multiple segments. The global AI assistant market is projected to grow from $3.35B in 2025 to $21.1B by 2030, a 44.5% CAGR. The broader AI companion market -- which includes personal memory and relationship-based AI -- is even larger at $37.7B, projected to reach $435B by 2034.

| SEGMENT | 2025 SIZE | PROJECTED | CAGR |
|---|---|---|---|
| Global AI Assistant | $3.35B | $21.1B by 2030 | 44.5% |
| AI Companion (Global) | $37.7B | $435B by 2034 | 31.2% |
| India AI Market | $5.1B | $17B by 2027 | 3x growth |
| India CPaaS (WhatsApp) | $1.12B | $2.36B by 2031 | 13% |
| Assistive Technology (Global) | $26B | $40B by 2030 | 7.3% |
| India Drone Market | $1.5B | $4.4B by 2027 | 38% CAGR |

**MARKET INSIGHT**

OmniPilot sits at the intersection of three growing markets: AI assistants ($21B), assistive technology ($40B), and commercial drones ($4.4B India alone). The combined addressable market exceeds $65B by 2030. The strategy is to enter through the smallest market (personal AI companion), prove the technology, then expand into the higher-value hardware verticals.

## India: The Right Market



India presents a unique opportunity for OmniPilot that global competitors cannot replicate:

| 1.03B | 660M | 78% | 68% |
|---|---|---|---|
| INTERNET USERS | SMARTPHONES | SMBS ON WHATSAPP | SMBS USING AI |

### Why India Works for OmniPilot

| FACTOR | DATA POINT | IMPLICATION FOR OMNIPILOT |
|---|---|---|
| **WhatsApp Dominance** | 78% of Indian SMBs use WhatsApp for business | WhatsApp-first distribution. No app install barrier |
| **Device Readiness** | 65% of 4GB+ phones can run lightweight on-device AI | On-device Whisper runs on existing hardware |
| **AI Adoption** | 68% of SMBs already using AI tools | Market is educated. Not early-adopter risk |
| **Price Validation** | Rs 399-500/month proven by Google & OpenAI | Price ceiling is known. No pricing risk |
| **Government Backing** | Rs 1.17B IndiaAI Mission + ADIP scheme for assistive devices | Subsidized hardware distribution channel |
| **Language Gap** | 22 official languages, most AI products are English-only | Hindi/regional language support = massive moat |
| **Blind Population** | 60-70M visually impaired, no affordable AI device exists | Assistive hardware has zero competition under Rs 15K |

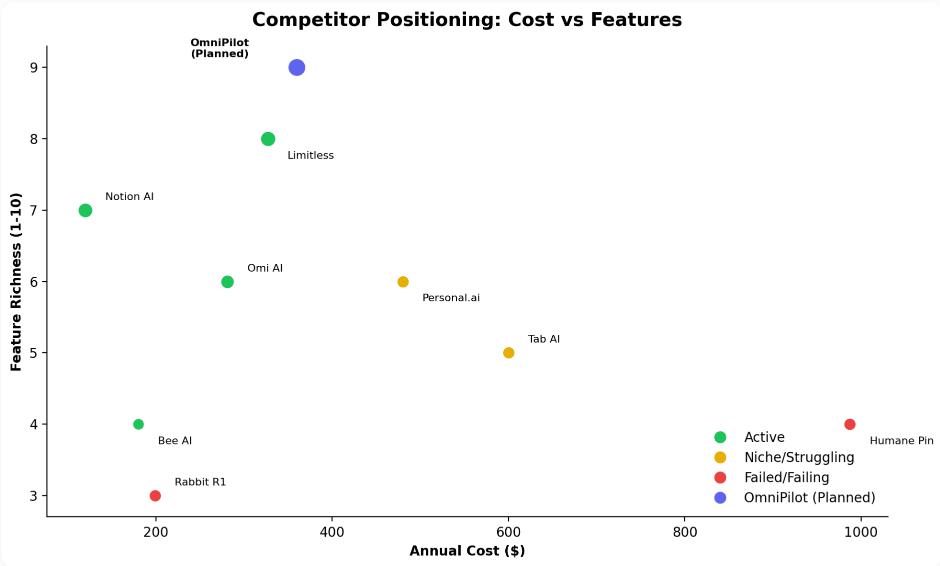
### Pricing Reality for India

| SEGMENT | SWEET SPOT | EVIDENCE |
|---|---|---|
| Consumer AI subscription | Rs 399-500/month ($5-6) | Google One AI, ChatGPT Plus pricing in India |
| SMB tools | Rs 1,000-2,000/month ($12-24) | Zoho, Freshworks SMB tiers |
| WhatsApp automation | Rs 8,000-15,000/month | WATI, Interakt, Gupshup business pricing |
| Assistive hardware | Rs 5,000-15,000 (ADIP covered) | Government scheme ceiling |
| Drone services | Rs 500-2,000/acre (agriculture) | Garuda, IoTechWorld market rates |

**PRICING CAUTION**

Above Rs 2,000/month, consumer adoption drops sharply outside enterprise. The B2C play must stay under Rs 500/month. The real revenue comes from B2B add-ons (BizBot, AdPilot, QuickBillPro integration at Rs 3K-8K/month) and hardware margins (blind device, pendant).

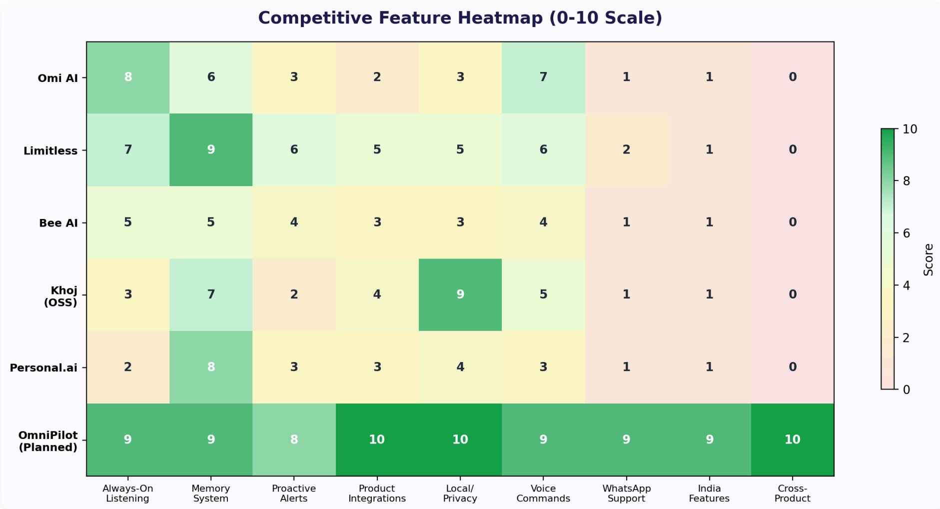
## Competitor Landscape




## Detailed Competitor Analysis

| PRODUCT | TYPE | PRICE | STATUS | KEY LEARNING FOR OMNIPILOT |
|---|---|---|---|---|
| **Omi AI** | Hardware pendant | $89 + $16/mo | Active | Cheap hardware works. Transcription accuracy is the weak point. Open-source firmware. BLE audio streaming to phone. Battery ~6hrs. Chinese manufacturing keeps BOM low. Community of 5,000+ users. |
| **Limitless** | Software + Pendant | $99 + $19/mo | Strongest player | Meeting intelligence is the beachhead feature. Desktop app came first, pendant second. Raised $13.5M Series A. Meeting prep, real-time notes, post-meeting summaries. The model to follow. |
| **Bee AI** | Software app | $15/mo | Active, niche | Software-first pivot was smart. Hard to differentiate. Life logging with AI summaries. Struggles with cold-start problem -- needs 2-3 weeks of data before useful. |
| **Tab AI** | Hardware pendant | $600 | Early, unproven | Price too high for pendant. Premium positioning may work for enterprise. Concept is good, execution unvalidated at scale. |
| **Rabbit R1** | Hardware device | $199 | 95% abandoned | Everything it does, a phone does better. Dedicated hardware for general computing is a dead end. LAM (Large Action Model) concept was ahead of execution. |
| **Humane AI Pin** | Hardware wearable | $699 + $24/mo | Dead (Feb 2025) | Cautionary tale. Laser projector gimmick. Overheated. 4-hour battery. Bricked when company shut down. Never be 10x worse than a phone. |
| **Khoj** | Open-source | Free / self-hosted | Growing | Open-source personal AI with memory. Good architecture reference. Lacks voice-first design and business integrations. |
| **Personal.ai** | Software | $40/mo | Slow adoption | Cold-start problem is real. Needs massive data ingestion before useful. Price too high for the value delivered early on. |

## Feature Comparison Heatmap




## What the Winners Got Right

| PATTERN | EVIDENCE |
|---|---|
| **Software-first beats hardware-first** | Limitless, Notion AI, Bee AI all healthy and growing. Humane, Rabbit failed catastrophically. Build the software, prove the value, then optionally add hardware. |
| **Meeting intelligence is the beachhead** | Every successful product started with "make meetings useful." Daily meetings are the universal pain point. Limitless, Otter, Fireflies all validated this. |
| **Sub-$100 hardware ceiling** | $89 Omi sells well. $199+ Rabbit struggles. $699 Humane dies. Consumer hardware for AI companions must be impulse-buy priced. |
| **Privacy is a selling point** | Users accept cloud IF they trust the company. Fully local is a growing niche. Apple's on-device AI push validates this direction. |
| **Phone is the real competitor** | Any dedicated device must be 10x better than a phone app at the specific task -- not 10x worse at everything. Pendant + phone is the winning combo. |

## What Failed and Why

| PRODUCT | ROOT CAUSE OF FAILURE | LESSON FOR OMNIPILOT |
|---|---|---|
| **Humane AI Pin** ($699) | Solved no problem better than a phone. Overheated within 20 minutes. Laser projector unusable in daylight. 4-hour battery. $24/month subscription for worse-than-Siri AI. Company shut down Feb 2025, bricking all devices. | Never build hardware that competes with phones on general tasks. Hardware must solve a specific problem phones cannot (always-on audio, assistive vision, drone control). |
| **Rabbit R1** ($199) | 95% user abandonment within 5 months. "Large Action Model" couldn't reliably complete tasks. Essentially a $199 Android app in a plastic box. | Software-first always. Phone is the platform. Dedicated hardware only for specific physical affordances (BLE pendant, camera for blind). |
| **Personal.ai** ($40/mo) | Cold-start problem: needs weeks of data before providing value. Users churned before reaching the "aha" moment. Price too high for early value. | Seed OmniPilot's memory from existing product data (BizBot chats, DevPilot sprints, TradePilot portfolio). Day-1 value, not week-3 value. |
| **Google Assistant** | Massive layoffs in 2024. AI assistant division restructured. Couldn't monetize consumer AI assistant. Pivoting to Gemini. | Consumer AI assistants need a revenue path beyond ads. B2B add-on pricing (BizBot, AdPilot) is the model. |

## OmniPilot's Five-Layer Competitive Moat

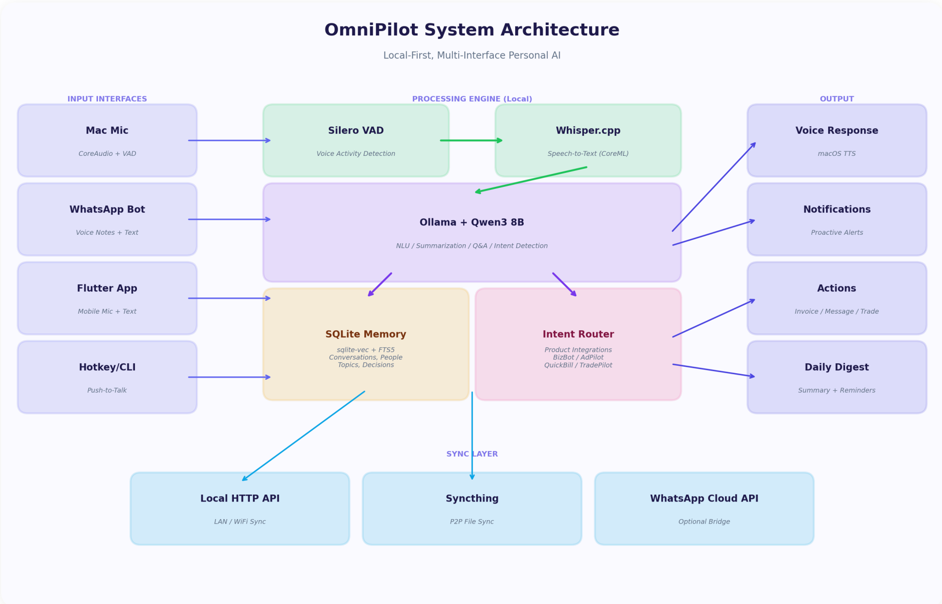
**MOAT ARCHITECTURE**

1. **Proprietary Product Graph** -- Deep integrations with 7 products spanning ads, billing, communication, trading, design, and development. No competitor has this.
2. **Shared Context Across Domains** -- Knows your invoice AND your campaign AND your trade position. Context compounds with every interaction.
3. **Indian Market Depth** -- GST compliance, WhatsApp-first, INR trading, Indian city services, Hindi voice support. Global competitors don't serve these.
4. **Existing Customer Base** -- BizBot, AdPilot, QuickBillPro already have paying customers. OmniPilot is an upsell, not a cold start.
5. **DevPilot Infrastructure** -- Battle-tested sprint/task/knowledge infrastructure with 43 MCP tools powers OmniPilot's backend orchestration.

PART 3

# Software Product

## System Architecture



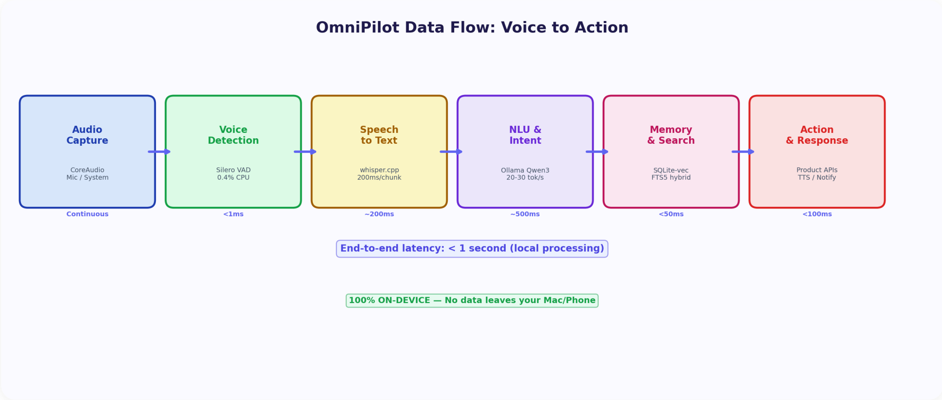

The OmniPilot software stack is a layered architecture designed for local-first operation. Every component runs on-device, with optional cloud connectivity only for features that require it (WhatsApp API, remote LLM fallback). The architecture is built around five layers:

| LAYER | COMPONENTS | RESPONSIBILITY |
|---|---|---|
| **Audio Input** | CoreAudio, AVAudioEngine, Silero VAD | Captures microphone audio, detects human speech using neural VAD (95% accuracy, 0.4% CPU), discards silence. Runs continuously as a menu bar service. |
| **Speech Processing** | whisper.cpp + CoreML, WhisperBridge | Converts detected speech segments to text using the small.en Whisper model (466MB). CoreML Metal acceleration delivers <500ms per chunk on M-series Macs. 27x real-time processing speed. |
| **Memory & Search** | SQLite, FTS5, sqlite-vec, all-MiniLM-L6-v2 | Stores transcripts with timestamps, people, topics, and decisions. Hybrid search combines FTS5 keyword matching with 384-dimensional vector similarity using Reciprocal Rank Fusion. |
| **Intelligence** | Ollama, Qwen3 8B Q4, IntentParser | Local LLM reasoning at 20-30 tokens/sec. Extracts action items, generates summaries, answers questions about past conversations. Intent parser routes commands to appropriate product APIs. |
| **Output** | AVSpeechSynthesizer, SwiftUI, Notifications | Indian English voice (Rishi) for spoken responses. 5-tab UI for browsing memories, tasks, and live transcription. macOS notifications for scheduled reminders. |

HOW TO THINK ABOUT IT

The architecture works like a human brain: the Audio layer is your ear (always listening), the Speech layer is your auditory cortex (converting sounds to words), the Memory layer is your hippocampus (storing and retrieving experiences), the Intelligence layer is your prefrontal cortex (reasoning and making decisions), and the Output layer is your voice and hands (responding and acting).

## Data Flow Pipeline



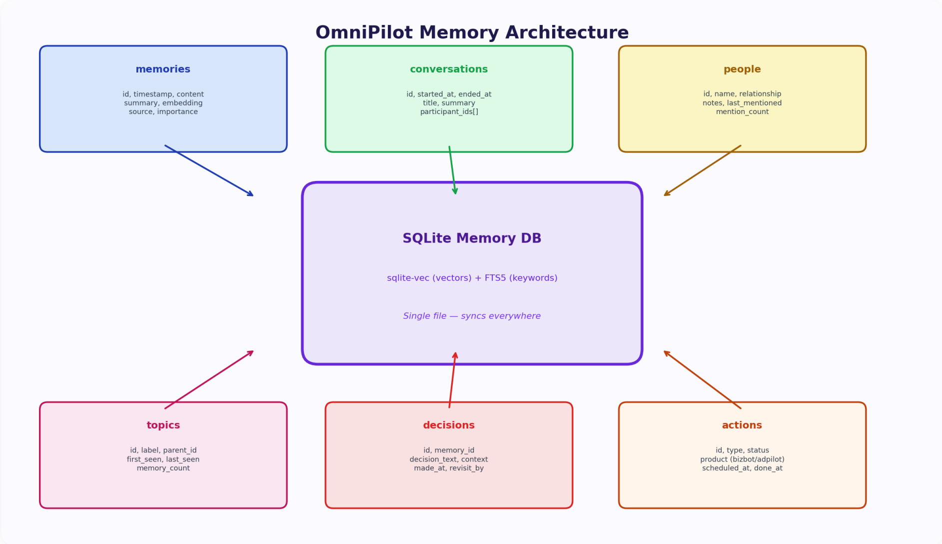

Every piece of audio goes through a precise pipeline from microphone to actionable output:

| # | STAGE | LATENCY | WHAT HAPPENS |
|---|---|---|---|
| 1 | **Audio Capture** | Real-time | CoreAudio captures microphone input as 16kHz PCM float32 frames. AVAudioEngine provides low-latency tap on the default input device. |
| 2 | **Voice Detection** | ~10ms | Silero VAD (ONNX neural network) analyzes each 30ms frame. Returns probability of speech. Threshold: 0.5. Below threshold = silence, discarded immediately. Saves 90%+ of processing. |
| 3 | **Chunking** | ~2s buffer | Detected speech is buffered into 2-5 second chunks. Chunk boundaries are set by 1.5s of silence or 30s max duration. Each chunk becomes a Whisper job. |
| 4 | **Transcription** | 200-500ms | whisper.cpp processes each chunk using CoreML Metal backend. small.en model (466MB) provides best accuracy/speed tradeoff. Output: text + timestamps + language confidence. |
| 5 | **Embedding** | ~50ms | all-MiniLM-L6-v2 generates a 384-dimensional embedding vector for the transcript. Used for semantic search alongside FTS5 keyword search. |
| 6 | **Memory Storage** | ~5ms | SQLite stores: raw text, embedding vector, timestamp, detected people/topics (extracted by LLM), source (mic/WhatsApp/manual), and metadata. |
| 7 | **Intent Parsing** | ~100ms | IntentParser checks if the transcript contains a command ("remind me", "send invoice", "what's my portfolio"). If yes, routes to TaskScheduler or product API. |
| 8 | **LLM Analysis** | 1-3s | Ollama Qwen3 8B processes the transcript for: action item extraction, topic classification, sentiment detection, summary generation. Runs asynchronously. |
| 9 | **Voice Output** | ~200ms | AVSpeechSynthesizer speaks the response using the Indian English "Rishi" voice. Rate: 0.5, pitch: 1.0. Only triggered for direct queries or scheduled reminders. |

**END-TO-END LATENCY**

From speech to stored memory: **<3 seconds**. From question to spoken answer: **<5 seconds**. All local. No network round-trips. Works in airplane mode.

## Memory Architecture




Memory is OmniPilot's core differentiator. Unlike stateless assistants (Siri, Alexa) that forget everything after each query, OmniPilot maintains a rich, searchable memory of every conversation, decision, and task.

### Database Schema

| TABLE | KEY COLUMNS | PURPOSE |
|---|---|---|
| **memories** | id, text, embedding, timestamp, source, people, topics | Core memory store. Every transcript, note, and observation is a memory. FTS5 index on text for keyword search. sqlite-vec index on embedding for semantic search. |
| **people** | id, name, relationship, last_seen, context | People mentioned in conversations. Auto-extracted by LLM. "Rahul from accounting" becomes a people entity linked to memories. |
| **topics** | id, name, category, frequency | Recurring topics (projects, clients, decisions). Used for daily summaries: "You discussed TradePilot 5 times today." |
| **tasks** | id, title, due_date, status, source_memory_id | Action items extracted from conversations or created by voice command. Linked to the memory that spawned them. |
| **summaries** | id, date, type, content | Daily, weekly, and topic summaries generated by the LLM. "This week you made 3 decisions about..." |

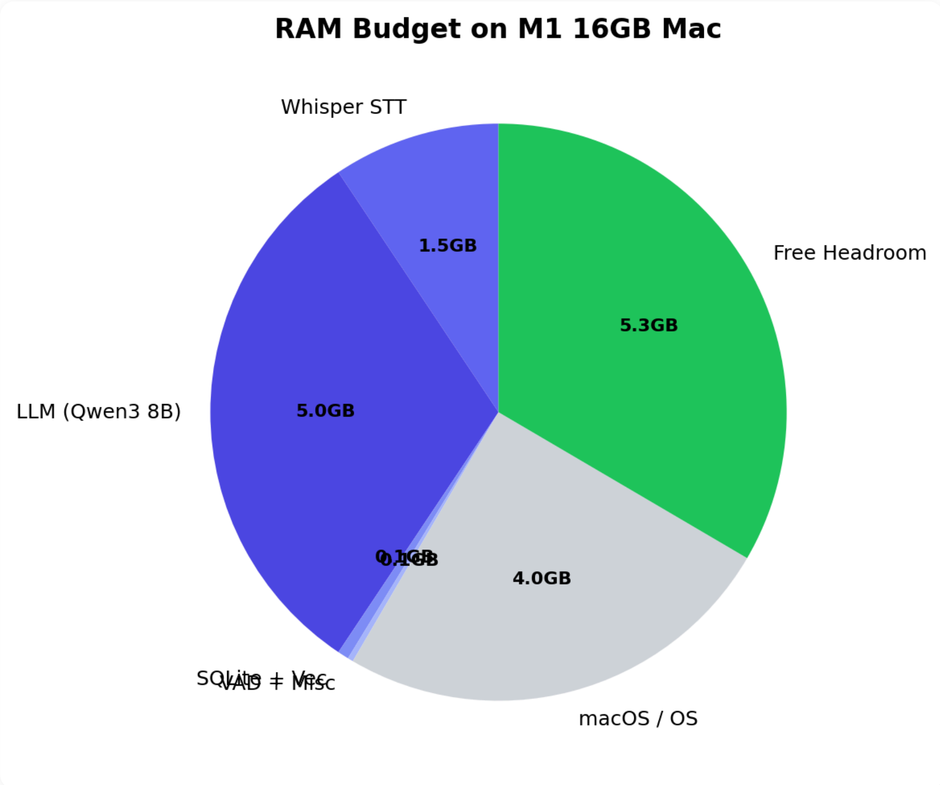
### Hybrid Search: How Recall Works

When you ask "What did I discuss with Rahul about the invoice?", two search systems fire simultaneously:

| SEARCH TYPE | ENGINE | HOW IT WORKS |
|---|---|---|
| **Keyword** | SQLite FTS5 | Matches exact words: "Rahul", "invoice". Fast, precise, handles typos poorly. Returns ranked list by BM25 relevance. |
| **Semantic** | sqlite-vec (384-dim) | Embeds query with MiniLM, finds memories with similar meaning. Catches "billing discussion" even if "invoice" wasn't said. Cosine similarity ranking. |
| **Fusion** | Reciprocal Rank Fusion | Merges both ranked lists. A memory ranked #2 in keyword and #3 in semantic scores higher than one ranked #1 in only one. Best of both worlds. |

> **WHY HYBRID SEARCH MATTERS**
>
> Pure keyword search misses context ("payment" vs "invoice" vs "billing"). Pure semantic search misses specifics (can't find exact names or numbers). Reciprocal Rank Fusion combines both, delivering 30-40% better recall than either alone. This is the same technique used by production search engines at Google and Bing.

## Tech Stack & RAM Budget




Everything runs on a standard M-series Mac with 16GB RAM. Here is the precise resource allocation:

| COMPONENT | MODEL / VERSION | RAM | CPU | NOTES |
|---|---|---|---|---|
| **Silero VAD** | v5, ONNX | 50MB | 0.4% | Always-on. Neural network. MIT license. No telemetry. 95% speech detection accuracy. |
| **whisper.cpp** | small.en + CoreML | 1.5GB | Burst | Only active during speech. CoreML Metal acceleration. 27x real-time. 466MB model file. |
| **Ollama (Qwen3 8B)** | Q4_K_M quantization | 5GB | Burst | Local LLM server. REST API. 20-30 tokens/sec on M1. Only active during analysis/query. |
| **Embedding Model** | all-MiniLM-L6-v2 | 80MB | <1% | ONNX runtime. 384-dimensional vectors. ~50ms per embedding. 80MB model file. |
| **SQLite + FTS5 + vec** | 3.45+ | 100MB | <1% | Single file database. Grows ~1MB per 1000 memories. FTS5 + sqlite-vec extensions. |
| **Swift App + UI** | SwiftUI | 50MB | <1% | Menu bar app. Minimal UI footprint. 5 tabs. |
| **Python Services** | 3.11+ | 200MB | <2% | Embedding service + VAD service + WhatsApp watcher. Three lightweight processes. |
| **Total** | -- | **~7GB** | **<5%** | Leaves 9GB free for macOS + other apps on a 16GB machine. |

## What's Built: v0.3 Desktop Companion

The working prototype includes a complete audio-to-intelligence pipeline:

### Audio Pipeline (3 Swift files)

- **AudioCapture.swift** -- CoreAudio microphone tap using AVAudioEngine. Captures 16kHz PCM float32 frames from the default input device. Provides real-time audio buffer to VAD.
- **SimpleVAD.swift** -- Silero VAD integration via ONNX Runtime. Processes 30ms frames, outputs speech probability. Configurable threshold (default 0.5). Manages speech/silence state machine with configurable padding.
- **WhisperBridge.swift** -- Bridges to whisper_server.sh (persistent whisper.cpp process). Sends audio chunks via local HTTP. Receives transcription results. Handles retries and server health checks.

### Intelligence Layer (4 Swift files)

- **OllamaClient.swift** -- REST client for Ollama API (localhost:11434). Supports streaming and non-streaming modes. Conversation analysis, summarization, and Q&A. Model: Qwen3 8B Q4.
- **IntentParser.swift** -- Natural language understanding for voice commands. Detects intents: create_task, query_memory, set_reminder, product_command. Extracts entities: time, person, amount, product.
- **TaskScheduler.swift** -- Cron-like scheduler for voice reminders. Parses natural language times ("remind me at 3pm", "every Monday at 9am"). Fires AVSpeechSynthesizer at scheduled time.
- **TaskStore.swift** -- SQLite persistence for tasks. Status tracking (pending/done/overdue). Links to source memory that created the task.

### Memory & Output (3 Swift files + 1 Python service)

- **MemoryStore.swift** -- Core memory database. SQLite + FTS5 for keyword search + sqlite-vec for semantic search. Hybrid search with Reciprocal Rank Fusion. People and topic extraction.

- **VoiceOutput.swift** -- AVSpeechSynthesizer wrapper. Indian English voice (Rishi). Queued speech with interruption handling. Used for query responses and scheduled reminders.
- **NotificationHelper.swift** -- macOS notification integration. Task due notifications, summary ready alerts, system status messages.
- **embedding_service.py** -- Python microservice running all-MiniLM-L6-v2 via ONNX Runtime. HTTP endpoint for embedding generation. Called by MemoryStore for both indexing and search.

### Services Layer (3 Python/Shell files)

- **whisper_server.sh** -- Persistent whisper.cpp process. Keeps the model loaded in memory (avoids 2s cold-start per chunk). Accepts audio via HTTP POST, returns JSON transcription. Auto-restarts on crash.
- **vad_service.py** -- Silero VAD as a standalone Python service. Provides WebSocket endpoint for real-time speech detection. Used by the Swift app for low-latency VAD without ONNX Swift overhead.
- **whatsapp_watcher.py** -- Monitors a local directory for WhatsApp voice note exports. Transcribes using Whisper, stores in memory DB. Enables "voice note to memory" pipeline for mobile users.

### UI Layer (1 Swift file)

- **QueryView.swift** -- 5-tab SwiftUI interface: **Live** (real-time transcription feed), **Ask** (voice/text query with LLM response), **Tasks** (pending/completed task list), **Memories** (searchable memory browser with hybrid search), **Settings** (model selection, VAD sensitivity, voice output toggle).

## Sprint Progress

### Sprint 1: Desktop Companion MVP (OMNI-S1)

| TASK | TITLE | PRIORITY | STATUS |
|---|---|---|---|
| OMNI-001 | Audio capture pipeline (CoreAudio + AVAudioEngine) | Critical | Done |
| OMNI-002 | Silero VAD integration (ONNX, 0.4% CPU) | Critical | Done |
| OMNI-003 | Whisper bridge (whisper.cpp + CoreML) | Critical | Done |
| OMNI-004 | SQLite memory store (FTS5 + sqlite-vec) | Critical | Done |
| OMNI-005 | Ollama LLM client (Qwen3 8B) | High | Done |
| OMNI-006 | Intent parser & task scheduler | High | Done |
| OMNI-007 | Voice output (AVSpeechSynthesizer) | High | Done |
| OMNI-008 | 5-tab SwiftUI interface | Medium | Done |
| OMNI-009 | Embedding service (all-MiniLM-L6-v2) | High | Done |
| OMNI-010 | WhatsApp voice note processing | Medium | Prototype |
| OMNI-011 | Daily summary generation | Medium | In Progress |

| 9/11 | 82% | 14 | v0.3 |
|---|---|---|---|
| TASKS COMPLETE | SPRINT PROGRESS | SOURCE FILES | CURRENT VERSION |

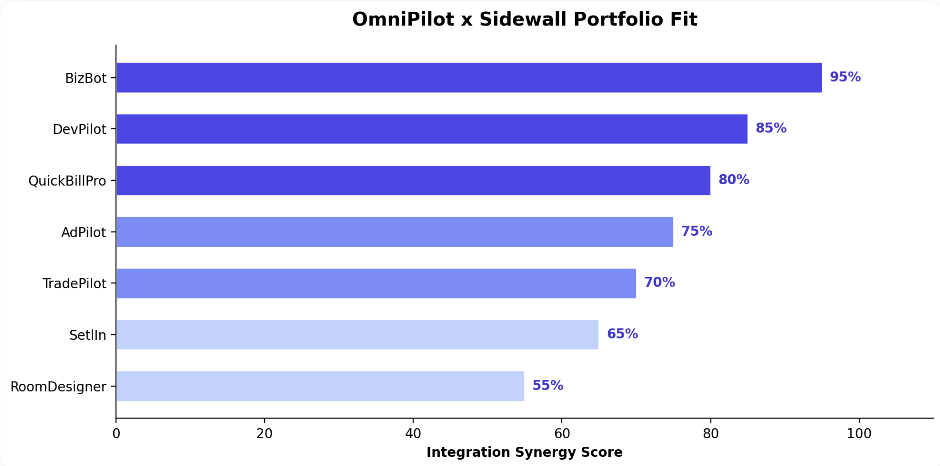
## Sprint 2: Product Integrations & Mobile (Planned)

| TASK | TITLE | PRIORITY | STATUS |
|---|---|---|---|
| OMNI-012 | BizBot integration -- voice commands for WhatsApp business automation | Critical | Planned |
| OMNI-013 | DevPilot integration -- voice-driven sprint/task management (43 MCP tools) | High | Planned |
| OMNI-014 | QuickBillPro integration -- voice invoicing and AR queries | High | Planned |
| OMNI-015 | Flutter mobile app -- on-device Whisper + background mic | Critical | Planned |
| OMNI-016 | Cross-product command routing (multi-product queries) | High | Planned |
| OMNI-017 | LAN sync (Mac hub + mobile clients via local HTTP API) | Medium | Planned |

PART 4

# Product Portfolio Integration

## Portfolio Fit Analysis




OmniPilot becomes the **unified voice interface** across the entire Sidewall product portfolio. Each integration creates a voice-first control layer over an existing business tool, turning multi-step workflows into single voice commands.

| 95% | 85% | 80% | 75% | 70% |
|---|---|---|---|---|
| BIZBOT SYNERGY | DEVPILOT | QUICKBILLPRO | ADPILOT | TRADEPILOT |

## Per-Product Integration Details

### BizBot (95% Synergy) -- Highest Priority

BizBot automates WhatsApp/Email/Call for businesses. OmniPilot becomes the voice layer on top.

| VOICE COMMAND | WHAT HAPPENS |
|---|---|
| "Reply to that caller -- tell them we'll send the quote by 5 PM" | OmniPilot routes response through BizBot's WhatsApp API. Composes message with context from the memory of the original conversation. |
| "What did Brand Tusker's last message say?" | Queries BizBot conversation history, reads the message aloud with AVSpeechSynthesizer. |
| "How many inquiries did we get today?" | "12 WhatsApp messages, 3 emails, 1 missed call. 2 need your reply." Aggregates across all BizBot channels. |
| *Overhears a phone call with a prospect* | Auto-creates a lead in BizBot CRM with context notes extracted from the conversation. |

**Revenue:** Rs 5K-8K/month add-on to existing BizBot subscription.

### DevPilot (85% Synergy)

| VOICE COMMAND | WHAT HAPPENS |
|---|---|
| "What's the sprint status for TradePilot?" | Queries DevPilot DB via MCP tools. Returns task counts by status. |
| "Mark task STAB-003 as done" | Voice-driven task update via MCP task_update tool. |
| *Developer says "same bug as P2P"* | Searches DevPilot learnings DB, surfaces the fix pattern from the P2P project. |

### QuickBillPro (80% Synergy)

| VOICE COMMAND | WHAT HAPPENS |
|---|---|
| "Send invoice to Brand Tusker -- Rs 2.3 lakhs plus GST" | Creates and sends GST-compliant invoice via QuickBillPro API. |
| "How much does client X owe me?" | Instant AR aging report without opening the app. |
| *Overhears "the payment came through"* | Auto-marks the relevant invoice as paid in QuickBillPro. |

### AdPilot (75% Synergy)

| VOICE COMMAND | WHAT HAPPENS |
|---|---|
| "What's my ROAS on the Brand Tusker campaign?" | Instant campaign metrics without opening dashboards. |
| "Pause underperforming campaigns" | Pauses all campaigns below configurable ROAS threshold. |
| "Alert me if any campaign CPC crosses Rs 15" | Sets up proactive monitoring with voice alerts. |

### TradePilot (70% Synergy)

| VOICE COMMAND | WHAT HAPPENS |
|---|---|
| "How's my portfolio today?" | Real-time P&L summary across all positions. |
| "What regime is Nifty in?" | Current market regime classification without opening the app. |
| *OmniPilot detects market news* | "Reliance dropped 4%. You have an open position." Proactive alerts. |

### SetlIn (65% Synergy) & DesignPilot (55% Synergy)

| PRODUCT | VOICE COMMAND | WHAT HAPPENS |
|---|---|---|
| SetlIn | "I need an electrician near Koramangala" | Triggers SetlIn's service recommendation engine for the user's neighborhood. |
| SetlIn | *Overhears "we just moved"* | Auto-triggers SetlIn onboarding flow for new residents. |
| DesignPilot | "Show me modern kitchen designs for 10x12" | Triggers DesignPilot room generation with specified dimensions and style. |
| DesignPilot | *While furniture shopping* | "Would this sofa fit my living room design?" Cross-references with DesignPilot room specs. |

## Cross-Product Intelligence (The Real Moat)

Single-product queries are useful. **Cross-product commands are transformational** -- and no competitor can replicate them:

> *"Brand Tusker's campaign is performing well. Send a thank-you on WhatsApp and draft next month's invoice."*
> *-- AdPilot + BizBot + QuickBillPro in one command*

> *"My TradePilot profits this month are Rs 50K. How much GST do I owe?"*
> *-- TradePilot + QuickBillPro tax calculation*

> *"What's the status across all my projects?"*
> *-- DevPilot sprints + AdPilot campaigns + TradePilot positions + BizBot conversations*

## Customer Journey

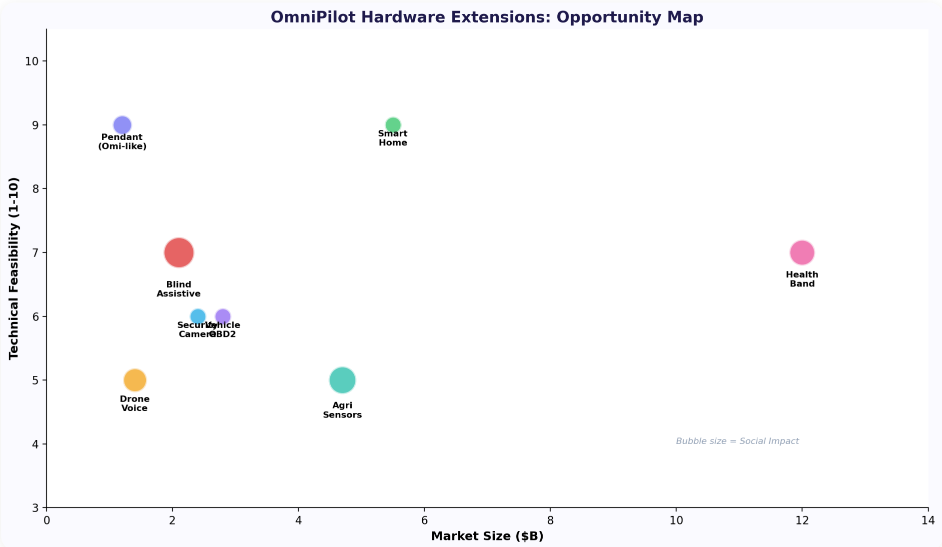

**THE COMPOUNDING EFFECT**

Each product integration adds context that makes every other integration smarter. BizBot tells OmniPilot about your clients. QuickBillPro tells it about their payment history. AdPilot tells it about campaign performance. Together, OmniPilot can say: "Brand Tusker hasn't paid their last invoice, their campaign is overspending, and they messaged on WhatsApp today -- here's what to do." No standalone AI assistant can do this.

PART 5

# Hardware Platform

## Hardware Opportunity Map




OmniPilot's software pipeline (VAD + STT + LLM + Memory) is hardware-agnostic. The same intelligence layer powers three distinct hardware products, each targeting a different market with different economics:

| DEVICE | MARKET | TARGET PRICE | TAM (INDIA) | UNIQUE ADVANTAGE |
|---|---|---|---|---|
| **BLE Pendant** | Professionals, life-loggers | Rs 5,000-7,000 | $3-5B | Omi competitor with Sidewall portfolio integration. Same concept, better software moat. |
| **Blind Assistive Device** | 60-70M visually impaired Indians | Rs 5,000-25,000 | $1-2B | Only affordable AI vision device in India. Hindi scene description. ADIP scheme funded. |
| **Drone Voice Control** | Agriculture, survey, defense | Module: Rs 15-25K | $4.4B by 2027 | Hindi voice commands for drone operators. DGCA compliant. Edge processing. |

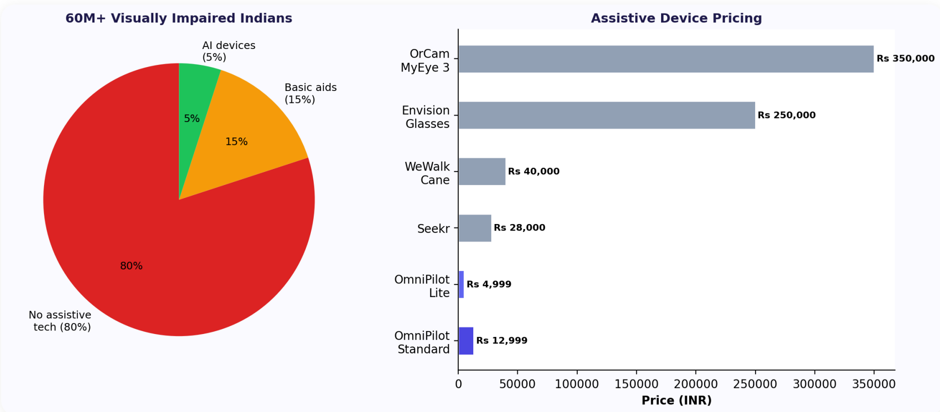
**PLATFORM THINKING**

Think of OmniPilot's AI pipeline like Android. Android runs on phones, tablets, watches, TVs, and cars -- same OS, different form factors. OmniPilot's pipeline (listen, understand, remember, act) runs on a Mac app, a BLE pendant, a vision device for blind users, and a drone controller. Same brain, different bodies. Each body opens a new market.

## Blind Assistive AI Device

India has 60-70 million visually impaired people. The most capable AI vision device (OrCam MyEye) costs Rs 3.5 lakhs. There is **nothing under Rs 15,000** that combines camera-based scene description with Indian language support. This is the gap OmniPilot fills.

## Market Opportunity




| 60-70M | 5-8M | Rs 3.5L+ | Rs 15K |
|---|---|---|---|
| VISUALLY IMPAIRED | TOTALLY BLIND | ORCAM PRICE | ADIP CEILING |

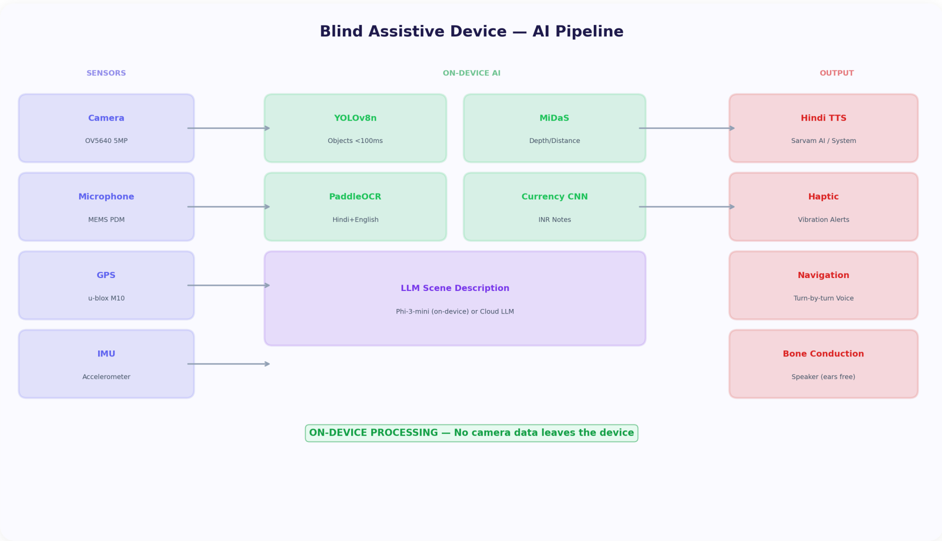
| METRIC | NUMBER | SOURCE & SIGNIFICANCE |
|---|---|---|
| Totally blind (India) | 5-8 million | WHO / NPCBVI estimates. Primary target for full vision assistance. |
| Moderate to severe impairment | 60-70 million | Lancet Global Health. Secondary market for navigation and OCR features. |
| Children blind (India) | 240,000+ | NPCBVI. Education use case -- reading textbooks, classroom navigation. |
| Low/middle income | ~80% of blind population | IHOPE Journal. Affordability is the #1 barrier, not awareness. |
| Blindness prevalence (India) | 0.36% (vs 0.2% global avg) | Census + surveys. India has disproportionately high blindness rates. |

## The Language Gap (Critical Differentiator)

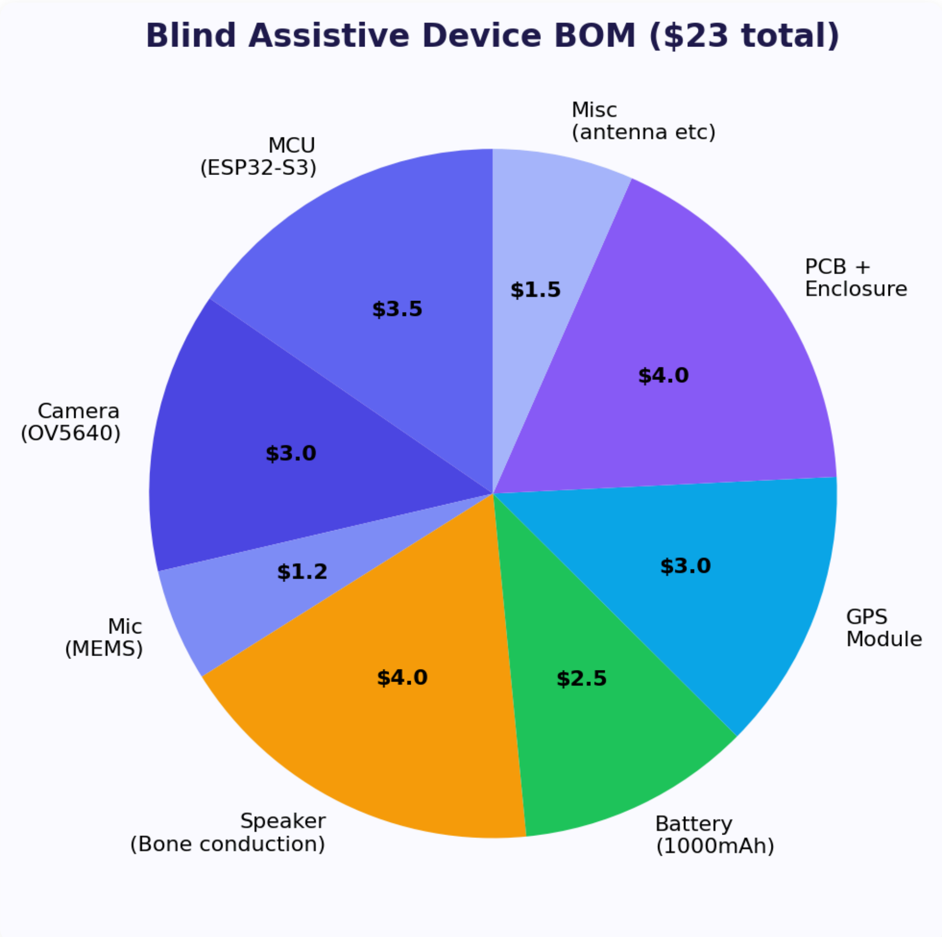
**NO PRODUCT NAILS INDIAN LANGUAGES**

India has 22 official languages. Current state: Hindi OCR accuracy is 70-80% (vs 99% English). Tamil, Telugu, Kannada OCR drops to 50-65%. Scene description in Indian languages is virtually non-existent in any product. Voice command in Hindi works on Google Assistant but no assistive device supports it natively. **A device that speaks Hindi fluently while describing the world = massive unmet need.**

## Device Architecture




## Bill of Materials



| COMPONENT | OPTIONS | UNIT COST | NOTES |
|---|---|---|---|
| **Processor** | Raspberry Pi Zero 2W / ESP32-S3 / Qualcomm QCS6490 | $15-80 | QCS6490 for production; Pi Zero for prototype |
| **Camera** | OV5647 (5MP) / IMX219 (8MP) wide-angle | $5-25 | 120-degree FoV pointing forward |
| **Microphone** | MEMS mic array (2-4 mics) | $2-5 | Noise cancellation for outdoor use |
| **Audio Output** | Bone conduction transducer / BLE earpiece | $5-15 | Ears remain open for safety (traffic awareness) |
| **GPS + Compass** | u-blox NEO-M8N + magnetometer | $10-15 | Navigation and heading for route guidance |
| **Battery** | 1000-2000mAh LiPo | $3-8 | Target: 4-8 hours runtime with duty cycling |
| **Connectivity** | WiFi + BLE (ESP32) or 4G LTE | $5-20 | BLE pairs with phone; 4G for standalone operation |
| **Enclosure** | 3D printed pendant, ~40g target | $2-5 | Neck pendant or lapel clip form factor |
| **Haptic Motor** | LRA vibration motor | $1-3 | Obstacle alerts via vibration patterns |
| **Total BOM (prototype)** | | **$50-175** | At scale (10K units): $60-80 = Rs 5,000-6,500 |

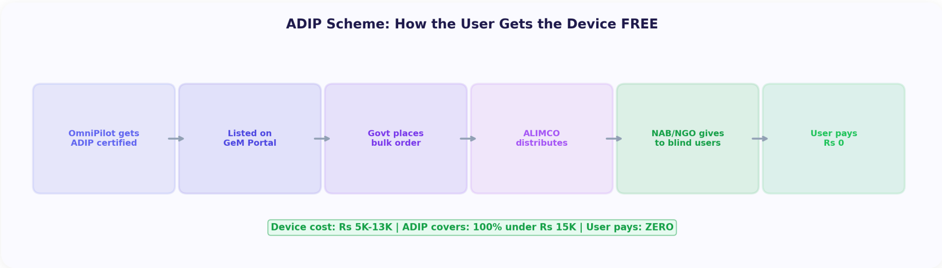
## Core Use Cases (Priority Order)

| # | USE CASE | TECHNOLOGY | ONLINE/OFFLINE | IMPACT |
|---|---|---|---|---|
| 1 | **Obstacle Alert** | MiDaS depth estimation + haptic | Offline | Head-height obstacles (branches, signs) that canes miss |
| 2 | **Currency ID** | Custom YOLOv8 (Indian notes) | Offline | Identifies Rs 10-2000 notes. RBI MANI only does this via phone |
| 3 | **Text Reading** | PaddleOCR + Piper TTS | Offline | Signs, menus, labels, letters -- in Hindi and English |
| 4 | **Scene Description** | GPT-4o / Claude multimodal | Online | "What's around me?" -- detailed environment description |
| 5 | **Face Recognition** | ArcFace-MobileNet + local DB | Offline | "Who is this person?" -- recognizes registered faces |
| 6 | **Navigation** | GPS + Google Maps API | Online | Turn-by-turn voice + haptic guidance to destinations |
| 7 | **Voice Q&A** | LLM multimodal | Online | "What color is this shirt?" -- visual questions answered |
| 8 | **Document Reading** | PaddleOCR (Hindi Devanagari) | Offline | Government forms, letters, prescriptions in Hindi |
| 9 | **Product ID** | ZXing barcode / ML Kit | Offline | Scan barcodes and QR codes at stores |
| 10 | **Emergency SOS** | Accelerometer + phone call | Online | Fall detection triggers emergency call to preset contacts |

## Competitive Comparison

| DIMENSION | ORCAM ($4,250) | ENVISION ($3,500) | WEWALK ($850) | OMNIPILOT VISION |
|---|---|---|---|---|
| **Price** | Rs 3.5L+ | Rs 2.9L+ | Rs 70K+ | **Rs 5K-15K** |
| **Always-on vision** | No (button press) | No (gesture) | No camera | **Yes (continuous)** |
| **Hindi scene description** | No | No | No | **Yes** |
| **Navigation** | No | No | Yes (voice) | **Yes (voice + haptic)** |
| **Indian currency** | Limited | No | No | **All denominations** |
| **ADIP eligible** | No (too expensive) | No | Borderline | **Yes (<Rs 15K)** |
| **Offline core** | Yes | Partial | Yes | **Yes** |
| **Form factor** | Glasses clip | Smart glasses | Cane handle | **Pendant/clip** |

## ADIP Scheme: Government-Funded Distribution




| SCHEME | COVERAGE | MAX AMOUNT | OMNIPILOT TIER |
|---|---|---|---|
| **ADIP (DEPwD)** | Free aids for disability >40%, income <Rs 30K/month | Rs 15,000 (100%) | Lite + Standard |
| **ADIP Extended** | 50% subsidy for Rs 15K-30K devices | Rs 30,000 (50%) | Pro tier |
| **NHFDC Loans** | Concessional loans for assistive devices | Up to Rs 10 lakh | Institutional bulk |
| **State Welfare** | Maharashtra, Tamil Nadu, Karnataka active programs | Varies | Regional distribution |

## Product Pricing Tiers

| TIER | PRICE | FEATURES | TARGET |
|---|---|---|---|
| **OmniPilot Lite** | Rs 4,999 | Obstacle alert + currency ID + basic OCR (offline only) | ADIP 100% covered |
| **OmniPilot Standard** | Rs 12,999 | All offline features + cloud scene description + navigation | ADIP 100% covered |
| **OmniPilot Pro** | Rs 24,999 | Everything + face recognition + continuous narration + Hindi TTS | ADIP 50% subsidy |

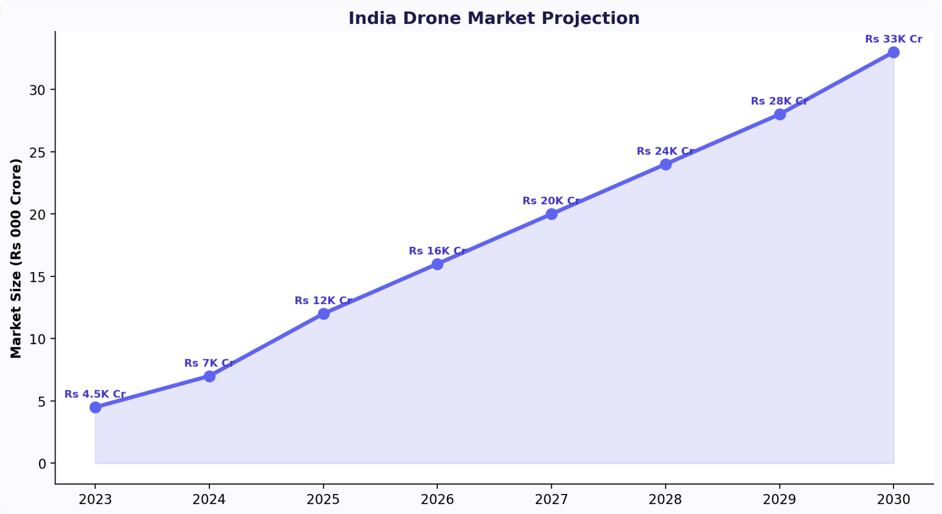
Cloud features subscription: Rs 99/month or Rs 999/year (subsidized through NGO partnerships).

> **THE HINDI MOAT**
>
> A device that speaks Hindi fluently while describing the visual world has zero competition globally. Sarvam AI's open-source Hindi models (TTS + STT) combined with PaddleOCR's Devanagari support and multimodal LLMs for scene description create a stack that no international competitor is building. This is OmniPilot's deepest hardware moat.

## Drone Voice Control System

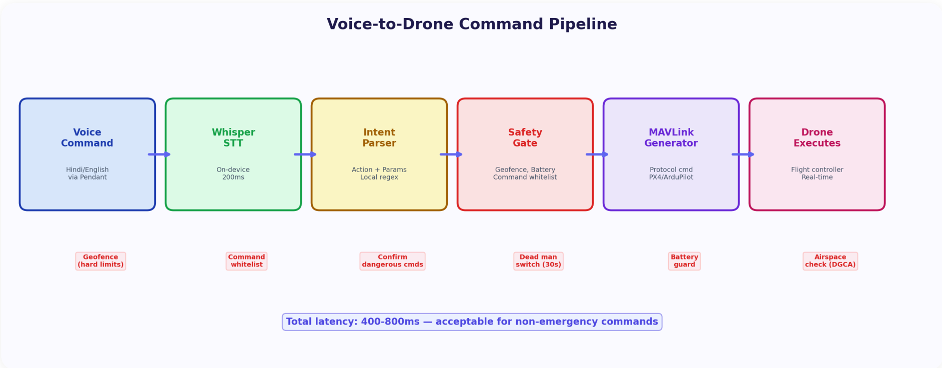
India's drone market is projected to reach $4.4B by 2027 (38% CAGR). Agricultural drones alone represent 65% of the market. Current drone interfaces require complex controllers and app-based flight planning -- a barrier for farmers and field operators. Voice-controlled drones with Hindi commands are a game-changer.

## Market Opportunity




| $4.4B | 38% | 65% | 1M+ |
|---|---|---|---|
| INDIA DRONE MARKET 2027 | CAGR | AGRICULTURE SHARE | DRONE PILOTS NEEDED |

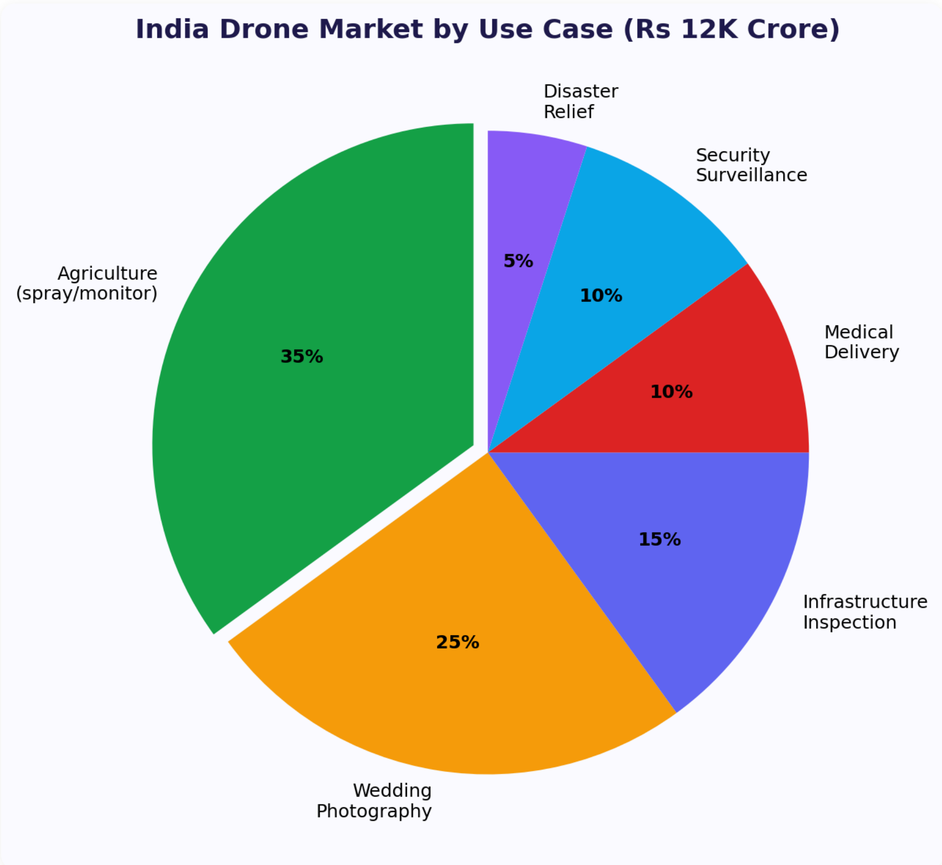
## Voice Control Pipeline




The drone voice control system reuses OmniPilot's core pipeline (VAD + STT + NLU) with a domain-specific command layer:

| STAGE | TECHNOLOGY | WHAT HAPPENS |
|---|---|---|
| **Wake Word** | Picovoice Porcupine | "Hey Drone" or customizable. Always-on, <1% CPU. Prevents accidental commands. |
| **Voice Capture** | MEMS mic array + noise cancellation | Outdoor-grade audio capture. Wind noise filtering. Beamforming for voice isolation. |
| **STT** | Whisper (Hindi + English) | On-device transcription. Supports code-switching (Hindi-English mix common in India). |
| **NLU** | Domain-specific intent parser | Maps voice to drone commands: takeoff, land, hover, go_to, spray, scan, return_home. |
| **Safety Check** | 6-layer safety architecture | Every command passes through geofence, altitude, battery, weather, airspace, and human proximity checks. |
| **MAVLink Output** | DroneKit / pymavlink | Translates validated commands to MAVLink protocol for autopilot execution. |

## Use Cases




| SECTOR | VOICE COMMAND EXAMPLE | VALUE DELIVERED |
|---|---|---|
| **Agriculture** | "Spray field 3, row pattern, 10 meters high" | Farmer controls spraying without complex app. Hindi commands for rural operators. Rs 500-2000/acre service revenue. |
| **Survey** | "Map this area, 50 meter grid" | Land surveyors get aerial mapping with voice. No flight planning expertise needed. |
| **Emergency** | "Search that area, thermal camera on" | Disaster response teams direct search drones by voice. Hands-free operation in crisis. |
| **Infrastructure** | "Inspect tower, circle pattern" | Telecom/power line inspectors command drones without controllers. Safer and faster. |

## DGCA Regulatory Summary

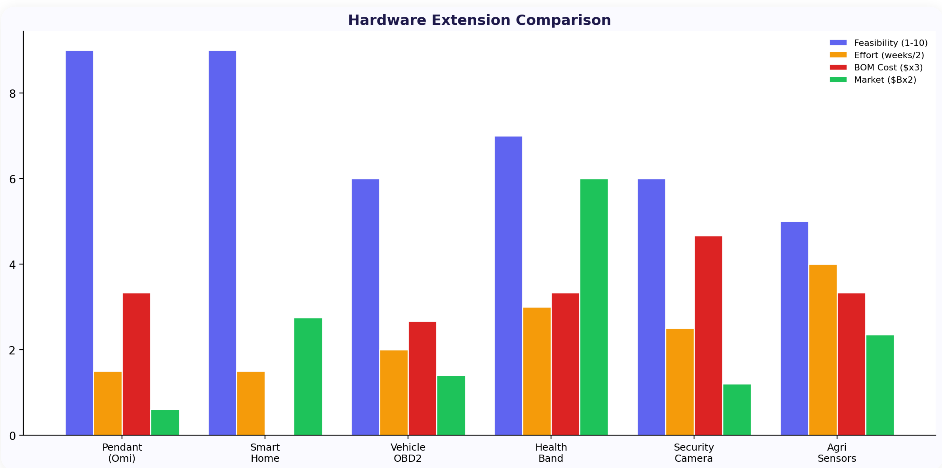
| REQUIREMENT | DETAIL | OMNIPILOT COMPLIANCE |
|---|---|---|
| **Registration** | All drones >250g must register on DigitalSky | Voice module is add-on; drone itself registers separately |
| **Pilot License** | Remote Pilot Certificate required for commercial ops | Voice control doesn't replace pilot; assists the pilot |
| **Geofencing** | No-fly zones enforced (airports, military, govt buildings) | Built-in geofence check before every command execution |
| **NPNT** | No Permission, No Takeoff -- digital unlock required | NPNT check integrated in safety layer |
| **Max Altitude** | 400 feet AGL for most operations | Altitude cap enforced in command validation |
| **Visual Line of Sight** | Pilot must maintain VLOS except with special permission | Voice control enhances VLOS ops (eyes on drone, not controller) |

## Safety Architecture (6 Layers)

| # | LAYER | CHECK | RESPONSE ON FAILURE |
|---|---|---|---|
| 1 | **Geofence** | Is the target within allowed airspace? | Reject command, voice warning: "Target is in restricted zone" |
| 2 | **Altitude** | Will the command exceed 400ft AGL? | Cap at maximum, voice confirmation of adjusted altitude |
| 3 | **Battery** | Enough power for command + return trip? | "Low battery -- returning home" if below 20% |
| 4 | **Weather** | Wind speed <35 km/h, no rain? | "Wind speed too high for safe operation" |
| 5 | **Airspace** | NOTAM check, temporary restrictions? | Block operation, explain restriction |
| 6 | **Proximity** | People/vehicles detected below? | "People detected in spray zone -- abort" |

## BLE Pendant (Omi Competitor)

The BLE pendant is the simplest hardware extension -- a Bluetooth audio capture device that pairs with the OmniPilot phone/Mac app. Unlike Omi ($89), the OmniPilot pendant's value comes from the Sidewall portfolio integration, not just transcription.



| SPEC | OMNIPILOT PENDANT | OMI AI | LIMITLESS PENDANT |
|---|---|---|---|
| **BOM** | Rs 500-800 ($6-10) | ~$15-20 | ~$25-30 |
| **Retail Price** | Rs 5,000-7,000 | $89 (~Rs 7,500) | $99 (~Rs 8,300) |
| **Audio** | BLE streaming to phone | BLE streaming to phone | BLE streaming to phone |
| **Battery** | 6-8 hours | ~6 hours | ~8 hours |
| **Unique Feature** | 7-product portfolio integration | Open-source firmware | Meeting intelligence |
| **India Support** | Hindi, WhatsApp, GST | English only | English only |
| **Software Moat** | Deep business tool integration | Generic transcription | Meeting-focused |

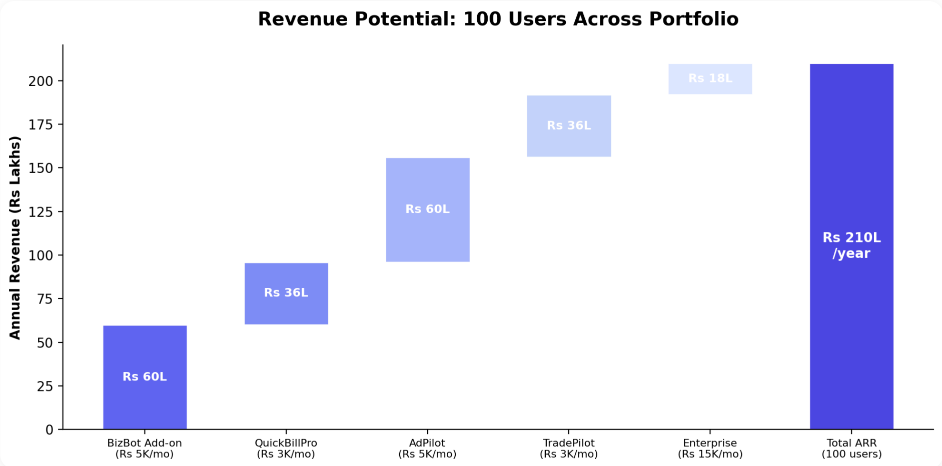
**PENDANT STRATEGY**

The pendant is a **distribution device**, not a profit center. Its purpose is to put always-on audio capture in users' hands so they generate more data for the memory system. The real revenue comes from the SaaS subscriptions and product integration add-ons that the pendant enables. BOM Rs 500-800, sell at Rs 5-7K, reinvest margin into software development.

PART 6

# Business Model

## Software Revenue Model




## Pricing Tiers

| TIER | INCLUDED | PRICE | TARGET CUSTOMER |
|---|---|---|---|
| **Personal** | Desktop companion, memory, daily summaries, 5 queries/day | Free | Personal use, dogfooding, community building |
| **Pro** | + WhatsApp bot + 2 product integrations + unlimited queries | Rs 499/mo | Individual professionals, freelancers |
| **Business** | + All 7 product integrations + cross-product commands + priority LLM | Rs 1,999/mo | SMB owners using Sidewall products |
| **Enterprise** | + Custom integrations + team features + dedicated support + SLA | Rs 15,000+/mo | Companies with 10+ users |

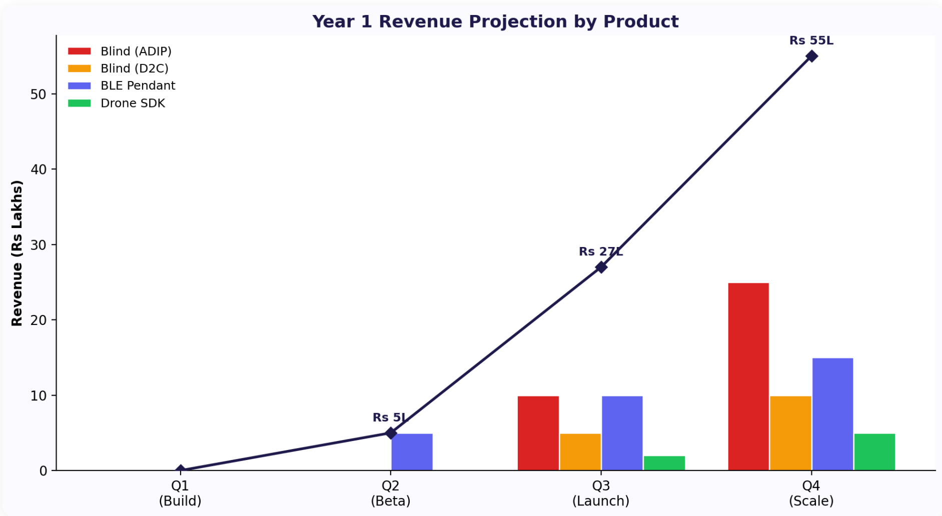
## B2B Add-On Revenue (Existing Customers)

| PRODUCT BASE | ADD-ON PRICE | VALUE PROPOSITION |
|---|---|---|
| **BizBot** | Rs 5K-8K/mo | "Manage BizBot with your voice -- reply to customers, check leads, send quotes, all hands-free" |
| **AdPilot** | Rs 3K-5K/mo | "Monitor campaigns, pause underperformers, get ROAS alerts -- all by voice" |
| **QuickBillPro** | Rs 3K-5K/mo | "Create invoices, check payments, track AR -- voice-first accounting" |
| **Enterprise Custom** | Rs 50K setup + Rs 15-25K/mo | Custom OmniPilot connected to company tools, workflows, and data |

## Unit Economics at 100 Users

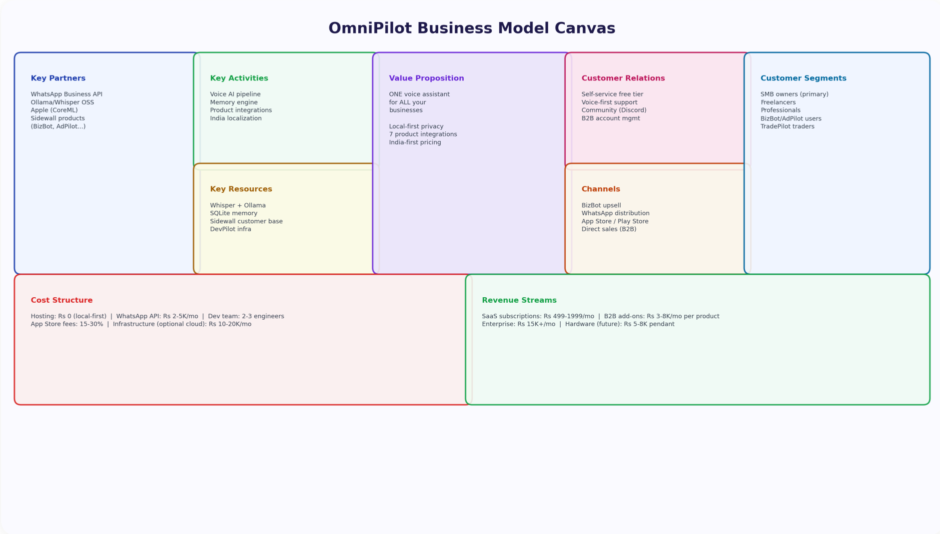
| REVENUE STREAM | MONTHLY | ANNUAL |
|---|---|---|
| BizBot add-on (100 users x Rs 5K) | Rs 5L | Rs 60L |
| QuickBillPro add-on (100 users x Rs 3K) | Rs 3L | Rs 36L |
| AdPilot add-on (100 users x Rs 5K) | Rs 5L | Rs 60L |
| TradePilot add-on (100 users x Rs 3K) | Rs 3L | Rs 36L |
| Enterprise (10 users x Rs 15K) | Rs 1.5L | Rs 18L |
| **Total ARR** | **Rs 17.5L/mo** | **Rs 2.1 Cr** |

## Hardware Revenue Model

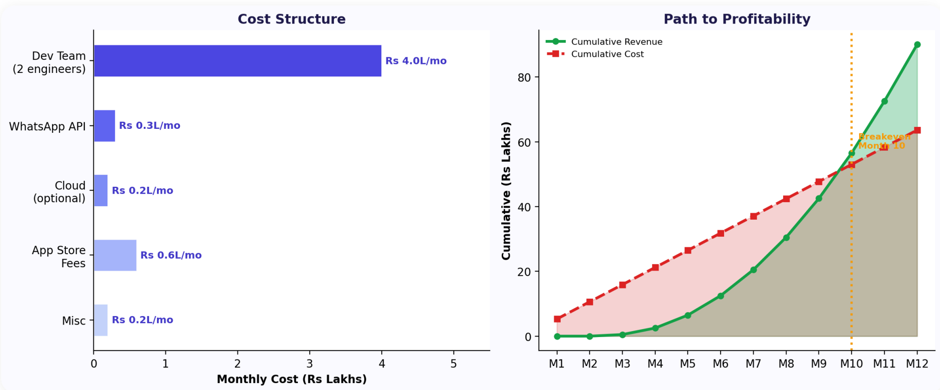



| DEVICE | RETAIL PRICE | BOM | MARGIN | YEAR 1 UNITS | YEAR 1 REVENUE |
|---|---|---|---|---|---|
| **BLE Pendant** | Rs 5-7K | Rs 500-800 | 85%+ | 500 | Rs 25-35L |
| **Blind Lite** | Rs 4,999 | Rs 3,000 | 40% | 1,000 | Rs 50L (ADIP funded) |
| **Blind Standard** | Rs 12,999 | Rs 6,000 | 54% | 500 | Rs 65L (ADIP funded) |
| **Blind Pro** | Rs 24,999 | Rs 10,000 | 60% | 200 | Rs 50L |
| **Drone Module** | Rs 15-25K | Rs 5-8K | 55-68% | 100 | Rs 15-25L |
| **Total Year 1 Hardware** | | | | **2,300** | **Rs 2-2.5 Cr** |

## Business Model Canvas

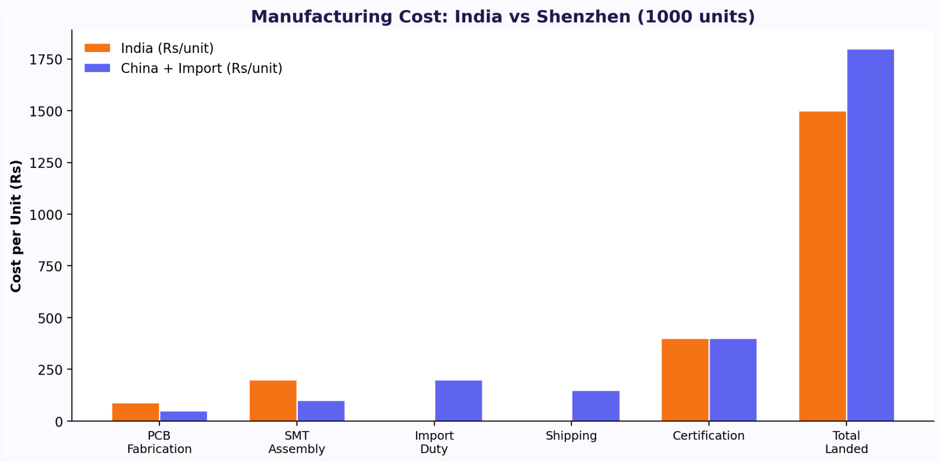



## Unit Economics Deep Dive

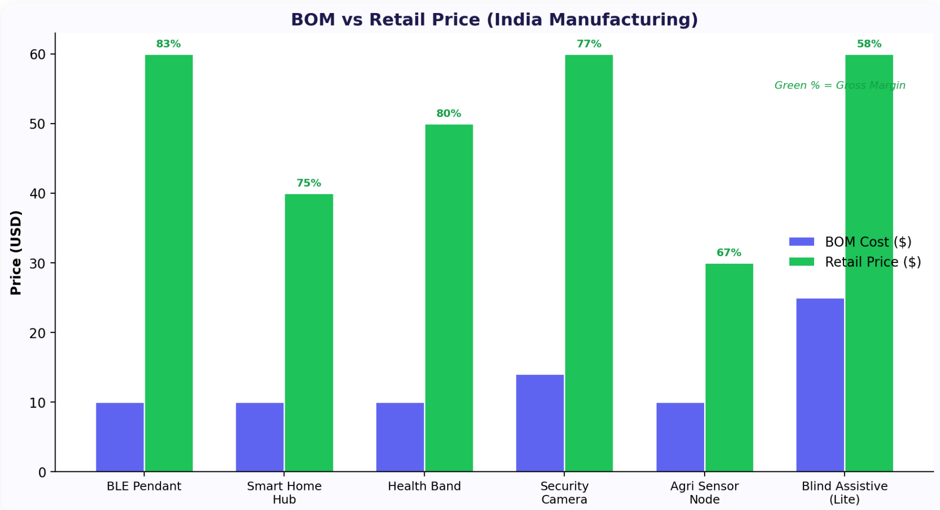



| METRIC | SOFTWARE | HARDWARE (BLIND) | HARDWARE (PENDANT) |
|---|---|---|---|
| Customer Acquisition Cost | Rs 500-1,500 | Rs 0 (ADIP/NGO) | Rs 1,000-2,000 |
| Avg Revenue/User/Month | Rs 3,000-5,000 | Rs 99 (cloud sub) | Rs 499-1,999 |
| Gross Margin | 90%+ | 40-60% | 85%+ |
| LTV (12 months) | Rs 36K-60K | Rs 6K-15K device + Rs 1.2K sub | Rs 6K device + Rs 6K-24K sub |
| LTV:CAC Ratio | 24-40x | Infinite (free acquisition) | 6-15x |
| Payback Period | 1 month | Immediate (device sale) | 1-2 months |

## Manufacturing Economics

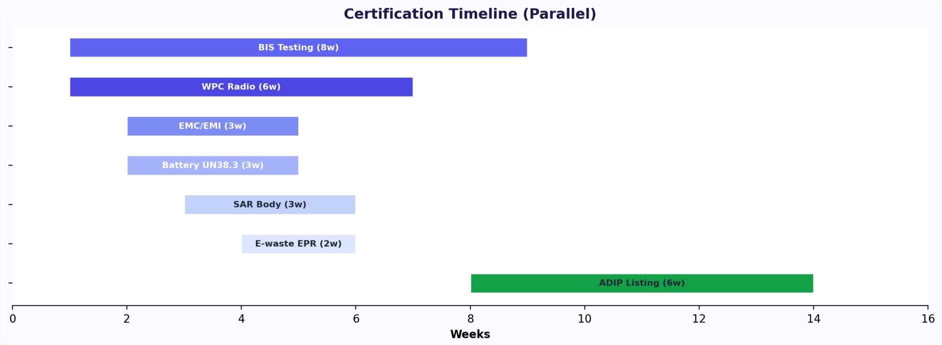



## BOM Comparison Across Devices




| FACTOR | INDIA MANUFACTURING | SHENZHEN MANUFACTURING |
|---|---|---|
| **BOM Cost** | 10-15% higher (import components) | Lowest BOM globally |
| **Assembly** | Rs 50-100/unit (manual) | $1-3/unit (automated) |
| **MOQ** | 500-1000 units | 1000-5000 units |
| **Lead Time** | 4-8 weeks | 6-12 weeks |
| **Quality Control** | On-site QC, easier iteration | Remote QC, higher defect risk |
| **Shipping** | Domestic delivery 2-5 days | Sea freight 30-45 days, customs |
| **Import Duty** | 0% (Make in India benefit) | 15-22% on electronics |
| **PLI Benefit** | 4-6% production incentive | Not applicable |
| **Best For** | Blind device (ADIP requires Indian manufacturing) | BLE pendant at scale (10K+) |

## Certification Timeline




| CERTIFICATION | TIMELINE | COST | REQUIRED FOR |
|---|---|---|---|
| **BIS (India)** | 6-12 weeks | Rs 1-3L | All electronics sold in India |
| **WPC** | 4-8 weeks | Rs 50K-1L | Wireless devices (BLE, WiFi) |
| **ADIP Scheme** | 8-16 weeks | Rs 50K | Government subsidy eligibility for blind device |
| **CE Mark** | 8-12 weeks | Rs 2-5L | Export to EU markets |
| **FCC** | 8-12 weeks | Rs 3-5L | Export to US market |
| **DGCA Type Cert** | 12-24 weeks | Rs 5-10L | Drone control module certification |

## Funding Sources

| SOURCE | AMOUNT | STAGE | FIT FOR OMNIPILOT |
|---|---|---|---|
| **BIRAC (DBT)** | Rs 50L-1Cr | Prototype to product | Blind assistive device (biotech/health-tech adjacent) |
| **DST-NIDHI** | Rs 25-50L | Seed / proof of concept | Technology innovation grants for startups |
| **Villgro (Social Enterprise)** | Rs 25-50L + mentoring | Early stage | Social impact focus (blind device) |
| **NASSCOM 10K Startups** | Mentoring + Rs 10L | Early stage | Product-market fit for SaaS + hardware |
| **IndiaAI Mission** | Rs 50L-5Cr | Growth | AI innovation for Indian market |
| **Assistive Tech VCs** | $500K-2M | Series A | Impact investing for blind device |
| **Drone Sector VCs** | $1-5M | Series A | AgriTech/drone sector focus (ideaForge, Garuda investors) |

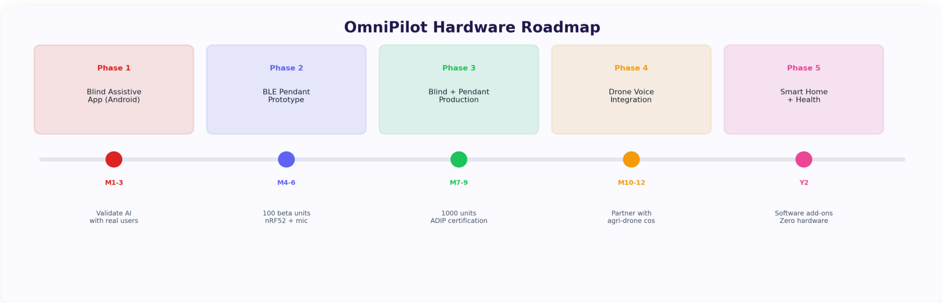
PART 7

# Execution Roadmap

## Software Roadmap


| PHASE | TIMELINE | DELIVERABLES | SUCCESS CRITERIA |
|---|---|---|---|
| **Phase 1** | Week 1-2 | Desktop companion (Mac menu bar, Whisper + Ollama + SQLite) | "Listen, remember, answer" working end-to-end. <5s query response. |
| **Phase 2** | Week 3-4 | WhatsApp bot (voice note processing, same memory DB) | Voice notes auto-transcribed and searchable. Daily summaries via WhatsApp. |
| **Phase 3** | Month 2 | Flutter mobile app (on-device STT, background mic) | Android app with continuous listening. Syncs with Mac via LAN. |
| **Phase 4** | Month 2-3 | BizBot integration (voice-to-WhatsApp, CRM queries) | First product integration live. 10 beta users. |
| **Phase 5** | Month 3-4 | Remaining product integrations (DevPilot, QuickBillPro, AdPilot, TradePilot) | Cross-product commands working. Full portfolio connected. |
| **Phase 6** | Month 4+ | Proactive intelligence (alerts, pattern detection, daily briefings) | OmniPilot initiates 5+ useful alerts/day without being asked. |

## Hardware Roadmap



| PHASE | TIMELINE | DELIVERABLES | SUCCESS CRITERIA |
|---|---|---|---|
| **H1** | Month 4-6 | BLE pendant prototype (nRF52840 + MEMS mic + BLE audio streaming) | 6-hour battery, stable BLE connection, works with OmniPilot app. |
| **H2** | Month 5-8 | Blind assistive software MVP (Android app with camera AI features) | OCR Hindi 80%+, currency detection 95%+, scene description working. |
| **H3** | Month 7-10 | Blind device hardware prototype (Pi Zero 2W + camera + bone conduction) | 50 beta testers across 3 cities. 8-hour battery. |
| **H4** | Month 8-12 | Drone voice module prototype + DGCA compliance | Voice commands reliable outdoors. Safety architecture validated. |
| **H5** | Month 10-14 | Custom PCB for blind device + BIS/ADIP certification | Rs 5K BOM at 1,000-unit run. ADIP scheme approved. |
| **H6** | Month 12-18 | Production manufacturing + distribution partnerships | 10,000 blind devices distributed via NGOs. 500 pendants sold. |

## 12-Month Combined Timeline

| MONTH | SOFTWARE MILESTONES | HARDWARE MILESTONES |
|---|---|---|
| **1** | Desktop companion v0.3 (done). WhatsApp processing prototype. | -- |
| **2** | WhatsApp bot live. Flutter app development starts. | BLE pendant component sourcing |
| **3** | Flutter app beta. BizBot integration live. | Pendant prototype assembly |
| **4** | DevPilot + QuickBillPro integrations. | Pendant beta testing (50 users). Blind app development starts. |
| **5** | AdPilot + TradePilot integrations. | Blind app MVP with camera AI features. Hindi OCR testing. |
| **6** | Cross-product commands. Proactive intelligence v1. | Blind app beta (20 users). Pendant BIS certification starts. |
| **7** | SetlIn + DesignPilot integrations. Full portfolio connected. | Blind hardware prototype (Pi Zero). Drone module design starts. |
| **8** | Performance optimization. Multi-language LLM (Hindi support). | Blind device 50-user pilot. Drone prototype indoor testing. |
| **9** | Team features (Enterprise tier). API for custom integrations. | Blind device 3-city pilot. Drone outdoor testing + DGCA prep. |
| **10** | Public launch (Pro + Business tiers). Marketing campaign. | Custom PCB design for blind device. Drone safety certification. |
| **11** | Scaling: 500+ users. Support infrastructure. | BIS/ADIP certification for blind device. Pendant mass production. |
| **12** | Rs 2Cr+ ARR target. Series A preparation. | 10K blind device manufacturing begins. NGO distribution agreements. |

## Budget Requirements

| CATEGORY | YEAR 1 BUDGET | KEY ITEMS |
| --- | --- | --- |
| **Software Development** | Rs 10-15L | Flutter developer, backend services, LLM fine-tuning, testing |
| **Hardware Prototyping** | Rs 5-8L | Component sourcing, 3D printing, PCB design, assembly (pendant + blind device) |
| **Certifications** | Rs 5-10L | BIS, WPC, ADIP scheme, DGCA type certification |
| **Manufacturing (Initial)** | Rs 10-15L | 1,000 blind devices + 500 pendants initial production run |
| **User Testing** | Rs 2-3L | 50 blind user pilot across 3 cities, usability studies |
| **Marketing & Distribution** | Rs 3-5L | NGO partnerships, digital marketing, conference presence |
| **Total Year 1: Rs 35-56L** | | Funded through: revenue (software), grants (blind device), bootstrapping |

**CAPITAL EFFICIENCY NOTE**

The software business (Rs 2.1Cr ARR at 100 users) can fund the hardware R&D (Rs 25-40L). The blind device qualifies for Rs 50L-1Cr in government grants (BIRAC, IndiaAI). The pendant has 85%+ gross margins from day one. This is not a VC-dependent burn model -- it's designed to be self-funding from software revenue + grants.

## Key Partnerships

| PARTNER | TYPE | ROLE IN OMNIPILOT |
| --- | --- | --- |
| **NAB India** | NGO | National Association for the Blind. Distribution channel for blind device. 1,000+ affiliated organizations. User testing access. |
| **Sightsavers** | NGO | International organization active in India. Funding for assistive tech pilots. Access to 15+ state programs. |
| **Garuda Aerospace** | Drone OEM | India's largest drone manufacturer. Integration partner for voice control module. Distribution to agricultural drone operators. |
| **VVDN Technologies** | EMS | Indian electronics manufacturer. PCB fabrication and assembly for blind device and pendant at scale. |
| **Sarvam AI** | AI/ML | Open-source Hindi language AI models. TTS and STT for Indian languages. Technical collaboration on Indic NLP. |
| **WhatsApp / Meta** | Platform | WhatsApp Cloud API for BizBot integration. 1,000 free conversations/month. Distribution channel for India. |
| **LV Prasad Eye Institute** | Medical | Clinical validation of blind device. User studies with ophthalmology patients. Credibility for ADIP scheme. |

## Success Metrics

| METRIC | 3 MONTHS | 6 MONTHS | 12 MONTHS |
|---|---|---|---|
| **Software Users** | 50 (beta) | 200 (Pro + Business) | 500+ |
| **Software MRR** | Rs 50K (beta testers) | Rs 5L | Rs 17.5L+ |
| **Product Integrations** | 2 (BizBot + DevPilot) | 5 | All 7 |
| **BLE Pendants Sold** | 0 | 50 (beta) | 500 |
| **Blind Devices (Pilot)** | 0 | 20 (3-city pilot) | 1,000+ (ADIP distribution) |
| **Drone Modules** | 0 | 0 | 100 (pilot customers) |
| **Hindi Accuracy (OCR)** | 70% | 80% | 90%+ |
| **Daily Active Memories** | 500/user | 2,000/user | 5,000+/user |
| **Cross-Product Queries** | 10/day | 100/day | 1,000+/day |

## The Flywheel

**THE COMPOUNDING LOOP**

More Sidewall products → More cross-product context → Smarter OmniPilot

↑

↓

More users ← Higher retention ← More value per query

Each new product in the Sidewall portfolio makes OmniPilot more valuable. Each new OmniPilot user makes the products stickier. Each hardware device (pendant, blind device, drone module) adds new data streams that make the AI smarter. This flywheel cannot be replicated by standalone AI assistant companies because they don't own the business tools that generate the context.

**THE AMAZON ANALOGY**

Amazon's flywheel: more sellers → more selection → better prices → more customers → more sellers. OmniPilot's flywheel: more products → more context → smarter AI → more users → more products. The key insight is the same: each investment compounds. BizBot makes OmniPilot smarter. OmniPilot makes BizBot stickier. The blind device generates grant funding that subsidizes software R&D. The pendant puts always-on audio in hands that generate data that makes the LLM smarter.

> *"The goal is not to build an AI assistant. The goal is to build the operating system for the Sidewall ecosystem -- a single intelligent layer that connects, remembers, and acts across every product, every device, every language." -- OmniPilot Vision Statement*

# OmniPilot

Personal AI Assistant + Hardware Platform
Software, Research & Hardware Strategy

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